Syllabus/ Orientation

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ABOUT MCNAIR SCHOLARS



"Whether or not you reach your goals in life depends entirely on how well you prepare for them and how badly you want them. You're eagles! Stretch your wings and fly to the sky."

"Before you can make a dream come true, you must first have one."

The Ronald E. McNair Postbaccalaureate Achievement Program awards grants to institutions of higher education for projects designed to prepare participants for doctoral studies through involvement in research and other scholarly activities. McNair participants are from disadvantaged backgrounds and have demonstrated strong academic potential. Institutions work closely with these participants through their undergraduate requirements, encourage their entrance into graduate programs, and track their progress to successful completion of advanced degrees. The goal of McNair is to increase the attainment of the Ph.D. by students from underrepresented segments of society.

Services provided by the program include

- Research opportunities for participants who have completed their sophomore year of college
- Mentoring
- Seminars and other scholarly activities designed to prepare students for doctoral studies
- Summer internships
- Tutoring
- Academic counseling
- Assistance in obtaining student financial aid
- Assistance in securing admission and financial aid for enrollment in graduate programs
- Fee waivers for graduate school applications and McNair Fellowships

The U of M's Ronald E. McNair Scholars Program is a TRiO program funded by the United States Department of Education with support from the U of M's College of Education and Human Development, the Office of the Vice Provost for Undergraduate Education and the Office of the Vice President for Diversity and Equity.

To Learn More…

University of Minnesota McNair Scholars http://www.cehd.umn.edu/trio/mcnair/
Federal TRiO Programs: http://www2.ed.gov/about/offices/list/ope/trio/index.html
U of M McNair Facebook: http://www.facebook.com/mcnairscholarsumntc

Biographical Data: Ronald E. McNair, Ph.D.

PERSONAL DATA: Born October 21, 1950, in Lake City, South Carolina. Died January 28, 1986. He is survived by his wife Cheryl, and two children. He was a 5th degree black belt Karate instructor and a performing jazz saxophonist. He also enjoyed running, boxing, football, playing cards, and cooking.

EDUCATION: Graduated from Carver High School, Lake City, South Carolina, in 1967; received a bachelor of science degree in Physics from North Carolina A&T State University in 1971 and a doctor of philosophy in Physics from Massachusetts Institute of Technology in 1976; presented an honorary doctorate of Laws from North Carolina A&T State University in 1978, an honorary doctorate of Science from Morris College in 1980, and an honorary doctorate of science from the University of South Carolina in 1984.

ORGANIZATIONS: Member of the American Association for the Advancement of Science, the American Optical Society, the American Physical Society (APS), the APS Committee on Minorities in Physics, the North Carolina School of Science and Mathematics Board of Trustees, the MIT Corporation Visiting Committee, Omega Psi Phi, and a visiting lecturer in Physics at Texas Southern University.

AWARDS: Posthumously awarded the Congressional Space Medal of Honor.

SPECIAL HONORS: Graduated magna cum laude from North Carolina A&T (1971); named a Presidential Scholar (1967-1971), a Ford Foundation Fellow (1971-1974), a National Fellowship Fund Fellow (1974-1975), a NATO Fellow (1975); winner of Omega Psi Phi Scholar of the Year Award (1975), Los Angeles Public School System's Service Commendation (1979), Distinguished Alumni Award (1979), National Society of Black Professional Engineers Distinguished National Scientist Award (1979), Friend of Freedom Award (1981), Who's Who Among Black Americans (1980), an AAU Karate Gold Medal (1976), five Regional Blackbelt Karate Championships, and numerous proclamations and achievement awards.

EXPERIENCE: While at Massachusetts Institute of Technology, Dr. McNair performed some of the earliest development of chemical HF/DF and high-pressure CO lasers. His later experiments and theoretical analysis on the interaction of intense CO_2 laser radiation with molecular gases provided new understandings and applications for highly excited polyatomic molecules.

In 1975, he studied laser physics with many authorities in the field at E'cole D'ete Theorique de Physique, Les Houches, France. He published several papers in the areas of lasers and molecular spectroscopy and gave many presentations in the United States and abroad.

Following graduation from MIT in 1976, he became a staff physicist with Hughes Research Laboratories in Malibu, California. His assignments included the development of lasers for isotope separation and photochemistry utilizing non-linear interactions in low-temperature liquids and optical pumping techniques. He also conducted research on electro-optic laser modulation for satellite-to-satellite space communications, the construction of ultra-fast

infrared detectors, ultraviolet atmospheric remote sensing, and the scientific foundations of the martial arts.

NASA EXPERIENCE: Selected as an astronaut candidate by NASA in January 1978, he completed a 1-year training and evaluation period in August 1979, qualifying him for assignment as a mission specialist astronaut on Space Shuttle flight crews.

He first flew as a mission specialist on STS 41-B which launched from Kennedy Space Center, Florida, on February 3, 1984. The crew included spacecraft commander, Mr. Vance Brand, the pilot, Commander Robert L. Gibson, and fellow mission specialists, Captain Bruce McCandless II, and Lt. Col. Robert L. Stewart. The flight accomplished the proper shuttle deployment of two Hughes 376 communications satellites, as well as the flight testing of rendezvous sensors and computer programs. This mission marked the first flight of the Manned Maneuvering Unit and the first use of the Canadian arm (operated by McNair) to position EVA crewman around *Challenger*'s payload bay. Included were the German SPAS-01 Satellite, acoustic levitation and chemical separation experiments, the Cinema 360 motion picture filming, five Getaway Specials, and numerous mid-deck experiments -- all of which Dr. McNair assumed primary responsibility. *Challenger* culminated in the first landing on the runway at Kennedy Space Center on February 11, 1984. With the completion of this flight, he logged a total of 191 hours in space.

Dr. McNair was assigned as a mission specialist on STS 51-L. Dr. McNair died on January 28, 1986 when the Space Shuttle *Challenger* exploded after launch from the Kennedy Space Center, Florida, also taking the lives of the spacecraft commander, Mr. F.R. Scobee, the pilot, Commander M.J. Smith (USN), mission specialists, Lieutenant Colonel E.S. Onizuka (USAF), and Dr. J.A. Resnik, and two civilian payload specialists, Mr. G.B. Jarvis and Mrs. S. C. McAuliffe.

For more information about the Ronald E. McNair:
http://abcnews.go.com/Technology/ronald-mcnair-challenger-explosion-victim-honored-hometown/story?id=12794042#.T5mqZDfNkxE

A Summary of the World

If we could shrink the Earth's population to a village of precisely 100 people, with all existing human ratios remaining the same, the world would look like this:

- There would be

  ___ Asians,
  ___ Europeans,
  ___ from the Western Hemisphere and
  ___ Africans (North and South).

- ___ would be non-white;
  ___ white.

- ___ would be non-Christians;
  ___ Christians.

- ____% of the entire world's wealth would be in the hands of only___ people

  Of those people, how many would be citizens of the United States
  _____

- ___ would be unable to read.

- ___ would suffer from malnutrition.

- ___ would live in sub-standard housing.

- ___ would have a college education.

- ___ would own a computer

When one considers our world from such an incredibly compressed perspective, the need for both tolerance and understanding become glaringly apparent.

Adapted from Phillip M Harter, MD, Stanford University, School of Medicine

Admissions Criteria Activity

The following represent admissions criteria for graduate schools. Please rank them in the order you think they would be ranked by most graduate departments. 1 for most important, 2 for second most important, etc…

_____ A. Recommendation letters from faculty known by members of the department.

_____ B. Particular sub scores on the appropriate GRE subject (advanced) test.

_____ C. Work experience.

_____ D. Impression made in personal interview.

_____ E. Personal Statement.

_____ F. Quality of undergraduate institution.

_____ G. Applicant known by your faculty.

_____ H. Recommendation letters from faculty not known by members of the department.

_____ I. Educational or career aspirations of applicant.

_____ J. Undergraduate grade point average overall.

_____ K. Undergraduate major related to intended field of graduate study.

_____ L. GRE General (Aptitude) Test quantitative score.

_____ M. GRE General (Aptitude) Test verbal score.

_____ N. Undergraduate grade point average in junior and senior years.

_____ O. Undergraduate grade point average in major field.

Self-Assessment Questionnaire (Pretest)

McNair Scholars Program

Please indicate your comfort level with aspects of the research process and applying to graduate school.

The Research Process	*Uncomfortable*	*Somewhat uncomfortable*	*Not sure*	*Somewhat comfortable*	*Very comfortable*
Writing skills (general)	1	2	3	4	5
Introduction/Hypothesis/ Statement of the problem/ Significance of study	1	2	3	4	
Literature review/ Citation styles	1	2	3	4	5
Research Method	1	2	3	4	5
Results/Discussion/ Implications	1	2	3	4	5
Abstract	1	2	3	4	5
Talking to and collaborating with a professor	1	2	3	4	5
Oral presentation skills	1	2	3	4	5
Preparing a Poster on your research	1	2	3	4	5
Graduate School Preparation					
Knowing that I want to go to graduate school	1	2	3	4	5
Preparing a curriculum vitae	1	2	3	4	5
Writing a personal statement	1	2	3	4	5
Finding a suitable graduate school programs	1	2	3	4	5
Finding funding for graduate school	1	2	3	4	5
Taking standardized test (e.g. ACT, GRE, LSAT)	1	2	3	4	5

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If you have begun to look at graduate schools, which ones are you interested in? What programs of study have you thought about (e.g. Public Policy)? Anything else you would like to share with McNair staff? All results are confidential.

McNair Scholars Mentor Profile

University of Minnesota

Assignment: Talk to your faculty mentor and write two paragraphs summarizing your findings. Paragraph One - Write a professional summary about your mentor using Section A as a guide. Paragraph Two - Write a summary of your mentor's graduate school experience and advice to you using Section B as a guide. *Please hand in these sheets along with your typed paragraphs on a separate page.*

Purpose: To get to know your mentor
To gain a further understanding about the graduate school process

Hint: In preparing for the discussion, review your mentor's faculty profile online (usually found on the department webpage under "Faculty") and print out a copy of your mentor's Curriculum Vitae (CV) if it is available. The CV will provide an outline which may help you better understand your mentor's story when you talk to him/her.

SECTION A: Professional Summary

Mentor's Name:

Mentor's Position at the U of M:

Received _____ (degree) in _____(field)

from _____(school).

Received _____ (degree) in _____(field)

from _____(school).

Received _____ (degree) in _____(field)

from _____(school).

Research Interests:

Publications:

Awards:

Title of the project you are working on together:

SECTION B: Graduate School

How did you decide where to do your Ph.D. (or highest degree)?

What has earning a Ph.D. (or highest degree) meant to you?

What was the most difficult aspect of completing your doctorate (or highest degree)?

What professional associations would benefit a student looking to attend graduate school in this field?

What are some names of journals/books that I might begin to explore?

What might I be doing right now to increase my chances of getting accepted into graduate school?

What graduate schools would you recommend I explore based on my research interests

Name:
SQ Access Code:

StrengthsQuest Assessment

This activity will take between 30 and 45 minutes and **should** be done by Monday, June 5. The McNair Seminar on Thursday, June 7, will focus on StrengthsQuest. Prior to starting the assessment you will be required to fill in contact information on the SQ site.

To complete part one of this activity go to this website:

www.strengthsquest.com

Use your University of Minnesota email and access code to log in; the access code that you will need is at the top of this page.

When you take the StrengthsQuest assessment, choose the items that **best** describe you and indicate to what extent they are descriptive of you. Gallup encourages you to indicate a preference on as many items as possible. If you choose the mid-point or "neutral" option on too many items, Gallup will not have enough data to provide you with your StrengthsQuest results.

To encourage you to answer with a "top of mind" response, the StrengthsQuest assessment is programmed to give you a specific amount of time to respond to each item. Try to respond within this time frame. Plan to take the inventory when and where you will not be interrupted.

Bring your list of talents to Week One and Two McNair Seminars, (either printed out or written down)! If you have problems or questions, contact Anthony Albecker at albecker@umn.edu.

In addition, be sure to complete the **SQ Verification** activity (See email attachment). This portion of the activity will take about 30 minutes.

Name: _____
Due Date: _____

Verifying Your Signature Themes

Directions: As you reflect on your Signature Themes and what it all means, think of two people who know you well and care about you to interview about your strengths. The idea here is to see your talents through their eyes. Your interviews may be done in person or over the telephone.

Interview # 1: Contact the first person. Explain that you are learning about your personal strengths through a scholarship program you've been selected for. Without revealing your SQ Talent Themes, ask them what they think 3-5 of your strengths are and to give an example of a time you demonstrated that strength. List the strengths and examples your interviewee shares with you below or on a separate sheet of paper or type them into a Microsoft Word document.

After you have the list, share with them what your SQ Talent Themes are. Note commonalities and differences between what they said and your SQ Talent Themes and discuss this with them. Any surprises? Takes notes on your discussion so you can share some highlights with the class.

STRENGTH	EXAMPLE
1.	
2.	
3.	
4.	
5.	

Interview #2: Have the **second** person read through the StrengthsQuest report (you can email your list of five talent themes to this person by going to see Strengths tab on the StrengthsQuest web site, and then go to Email a Friend), or read it to them and then ask the following questions. Record their answers below or on a separate sheet of paper or type them into a Microsoft Word document.

1. For each of your 5 Signature Themes ask:
   Do you see this theme in me? If so, please give me an example.

Do you see this theme in me?	Example
Theme 1:	
Theme 2:	
Theme 3:	
Theme 4:	
Theme 5:	

2. What surprises you about my signature themes?

3. Which of the five themes have you noticed most often in me?

4. Are there any additional talent themes that you see in me? What are those themes?

Research Timeline

McNair Scholars
University of Minnesota

Assignment: Talk to your faculty mentor, establish a timeline for your summer Research assignments/meetings, and create a detailed calendar or list of the research activities you will be engaging in during the 10-week program. Please include weeks 2 – 10 in your timeline.

Purpose: To organize your summer research schedule with your mentor and develop an understanding of your mentor's expectations for your work.

Sample Timelines (Calendar and List Styles):

Week 2					
	Monday June 10th	Tuesday June 11th	Wednesday June 12th	Thursday June 13th	Friday June 14th
McNair Assignments Due		Timeline, Mentor Profile, CV		8am Advisor Mtg	
Faculty Mentor's Assignments Due	Start reading assigned articles and making summaries	1:30 Take practice GRE, 110 ApH			NyTimes Article Summary Due
Meetings	9am Meet w/ Grad Asst in Wilson Library 310	10-12 McNair meeting	10am Mentor Meeting 170 HHH	3:30 Mentor Meeting 274 HHH	9-1pm Public Affairs Conference in St. Paul

SAMPLE

Monday June 10th
-9am Meeting with Grad Assistant in Wilson Library 310
-Start reading assigned articles and making summaries

Tuesday June 11th
-10am-12pm McNair Meeting in 40 ApH
-Timeline, Mentor Profile and CV due
-1:30pm Take practice GRE in 110 ApH

SAMPLE

Wednesday June 12th
-10am mentor meeting, 177 HHH

Thursday June 13th
-8am advisor meeting, 80A ApH
-3:30 mentor meeting, 274 HHH

Friday June 14th
-9am-1pm Public Affairs Conference in St. Paul
-NYTimes article summaries due at 3pm – email to mentor

RESUME v. CURRICULUM VITAE (called "CV" or "Vita")

Both resumes and CVs are marketing documents that provide key information about your skills, experiences, education, and personal qualities that show you as the ideal candidate. However, resumes and CVs differ in use, format, and length.

Resume

- ☑ Primarily for non-academic jobs
- ☑ Preferably one page, brevity important!
- ☑ Formatting/style important
- ☑ Customized to each job you apply for - can pick and choose which jobs to include
- ☑ Includes:
    - o Name/Contact Info
    - o Objective
    - o Education
    - o Work Experience
    - o Honors/Awards/Activities

CV

- ☑ Primarily for used for academic positions
    - o Graduate school apps
    - o Teaching jobs
    - o Research grants and fellowships
- ☑ As many pages as needed, completeness important! (3+ pages preferred)
- ☑ Formatting less important – should reflect the academic discipline you are entering
- ☑ Customize to each program you apply for and able to highlight relevant academic experiences
- ☑ Includes:
    - o Name/Contact Info
    - o Education
    - o Experience
        - ▪ Academic, Teaching, Reseach Positions
        - ▪ Non-academic Positions
    - o Honors and Awards
    - o Teaching or Research Interests
    - o Publications
    - o Presentations
    - o Conference Participation
    - o Committee or Professional Association Service
    - o Skills/Languages
    - o References (can be separate page)

HELPFUL HINTS FOR CREATING A CV

- ☑ New job seekers usually begin the Curriculum Vitae with academic preparation, which draws attention to the degree. An experienced job seeker, however, can begin with experience and place the educational preparation somewhere else.
- ☑ Prioritize your top skills or experience to be presented in the beginning sections of your CV.
- ☑ Distinguish between kinds of experience. For example, academic appointments, hospital or clinical, and/or professional experiences.
- ☑ Due to the length of the Vitae, preparing one that is not boring or that buries information can be a challenge.
- ☑ Do not include papers submitted or in preparation unless they have been accepted, in which case indicate "in press." Follow appropriate documentation guidelines for articles, books chapters, presentations, etc.
- ☑ Be specific when describing your experiences. For example: *Worked with employees in a residential setting*. Instead you may say: "Recruited, hired, trained and supervised more than 20 employees for a residential, for-profit agency." This is more likely to grab an employer's attention.
- ☑ Tailor your Vitae to each position you are applying for, whether it be an application to graduate school, or a full-time teaching position. Avoid attempting to create a one-size-fits-all Vitae to send to all employers. They expect your Vitae to state how you might fit a position with that organization.
- ☑ Consider your Vitae as a work in progress, updating it to better suit your current situation. Have more than one person review it.

WORDS THAT MAKE A DIFFERENCE When describing your experiences, action verbs should begin each statement. Try these:

achieved	established	maintained	streamlined
added	evaluated	managed	supervised
coordinated	expanded	organized	trained
created	generated	performed	utilized
designed	identified	planned	verified
developed	increased	reduced	worked
eliminated	initiated	saved	wrote

When reporting your accomplishments, use superlatives and other dynamic words. Here are a few:

first	only	most	highest	best
achieved	expanded	pioneered	completed	improved
reduced	resolved	restored	spearheaded	surpassed
transformed	won	overhauled	initiated	revitalized

Be concise and impressive! Don't be afraid to highlight your accomplishments!

Week 1

Starting Your Research Paper

As you write your research paper, please include the following sections: Abstract, Introduction, Hypothesis/Research Question, Statement of the Problem, Significance of the Study, Literature Review, Methodology, Results, Discussion, and Implications/Future Research. Remember that your research project is highly individualized and may not fit perfectly with this standard format for writing a research paper – please refer to your faculty research mentor in order to individualize this paper to your specific discipline (e.g., Sociology)

Tip: It is helpful to find examples of research writing in your discipline to help you gain an understanding of what your writing should look like. You can do this by looking at articles in a well-known journal in your field. For example, those researching in the field of psychology might look at the *American Journal of Psychology*.

Abstract

The abstract is the first section of your paper. It is essentially an executive summary of your paper and, therefore, is the very last section of the research paper that you will write. Do not worry about writing this section yet – it is not due until week 9 of our program.

Introduction

- Goal is to introduce the research project to your audience and draw the reader in (Start with a story!!)
- Gives context of the research
- Gives necessary background information
- Explains why your project is important (vaguely)
- Usually, the last sentence is the thesis statement (the main idea of your paper)
- Structure, length and content depend on discipline and research (and mentor)

1. Why do we need an introduction?
2. What do we want to include?
3. What should we not include?

Style considerations - The reader doesn't want to read everything you know about a subject. Present background information *only as needed* in order to support a position.

Hypothesis *or* Research Question

Ask your faculty mentor whether a *hypothesis* or *research question* is more appropriate for your field of research.

Hypothesis

What are you expecting to find? Your hypothesis will focus your ideas and information. It is the proposed answer to a research question, which you will finalize only after completing the research. (It's okay to modify and revise the working thesis as you research more about the topic or issue.) State the hypothesis precisely—do not oversimplify, and don't exaggerate. Your hypothesis should cover exactly the topic you want to talk about, no more and no less. Why commit yourself to an extreme opinion that you don't really believe in, and then look like you're contradicting yourself late on?

OR

Research Question

What is the question that underlies and drives forward your research?

Most social science research originates from some general problem or question. You might, for instance, be interested in what programs enable the unemployed to get jobs. Usually, the problem is broad enough that you could not hope to address it adequately in a single research study. Consequently, we typically narrow the problem down to a more specific **research question** that we can hope to address. The research question is often stated in the context of some theory that has been advanced to address the problem.

For instance, we might have the theory that ongoing support services are needed to assure that the newly employed remain employed. The research question is the central issue being addressed in the study and is often phrased in the language of theory. For instance, a research question might be: *Is a program of supported employment more effective (than no program at all) at keeping newly employed persons on the job?* (Retrieved from http://www.socialresearchmethods.net/kb/strucres.php on 5/27/08.)

It is important to have a good quality research question that you can actually answer during your 10 weeks of research with McNair Scholars. Make sure the question interests you enough to spark you own thoughts and opinions. Ask yourself, is the question easily and fully researchable (e.g., can I really research 30 online writing programs developed over a span of 10 years)? Is my question to broad, too narrow, or okay? What info will you need to answer the research question, and do you have access to those resources (journals, books, internet resources, government documents, people, etc.)?

Statement of the Problem

Although the introduction to a short paper may be only one paragraph in length, don't think of that as an ironclad rule. A good rule of thumb is that the introduction should be no longer than 15 percent of your paper. An introduction has two parts (posing the problem and addressing it); each major part has two or more subparts. The introduction ends with a statement of the thesis.

If you think that you've found a thesis, claim, conclusion – or at least a topic – try fitting it into one or more of the following:

1. Most people believe that _____, but a closer look will show that _____.
Writing in the university most often deals will saying something "new," something "against the grain." Although this sentence does not pose a problem per se, it can help you to arrive at a thesis which will resolve some misunderstanding. To help you (and your readers) figure out why clearing up this understanding is important, you will need one of the other steps.

2. What we know about _____ is that _____; what we don't know is _____.
Not all "new" knowledge is made by showing that someone else has "misunderstood." If you can find a "gap" in what is known, you've also taken the first step to posing a problem. As with #1, however, you need to use one of the other statements to figure out why this gap is worth filling in.

3. If we (do not) understand _____, we will (not) understand _____.
This tip is helpful especially when you are still at the topic stage – you know the subject you are writing about, but you don't yet know your thesis (what you want to say about it) or your organizational structure.

4. I am analyzing/comparing _____ so that I can explain/understand _____.
You have found a problem when you can say both what your paper will do (e.g., analyze) and how you and your readers will benefit from your having done it (the "so that" half). This step can work even if you are still searching for a thesis, as long as you know what you want the paper to do (e.g., analyze, classify, evaluate).

5. So what?
If you can answer some form of this question, you have found a "problem." This tip works especially well when you think you have a thesis.
(Adapted from Joseph Williams, The Little Red Schoolhouse, University of Chicago).

Retrieved on May 28, 2006, from http://www.temple.edu/writingctr/student_resources/statement_of_problem.htm.

Significance of the Study

- Goal is to articulate the importance of the study (be specific)
- Elaborate on the significance for researchers, policymakers, communities and specific populations
- Addresses the “gaps in the literature”, or areas that you cannot find information on when you conduct a literature review.
- In other words, what will your project contribute to existing literature/theory/policy?
- Why is your project unique?
- Structure, length and content depend on discipline and research (and mentor)
- This is the “So What” section. At the poster presentation and at the conferences, people will want to know why your project is significant.
- Many of you are working on projects that attempt to fill in areas that have not been researched, or have scarcely been researched.

Literature Review
(a.k.a. Lit Review or Review of the Literature)

1. What's a Literature Review?

Not to be confused with a book review, a **literature review** surveys scholarly articles, books and other sources (e.g. dissertations, conference proceedings) relevant to a particular issue, area of research, or theory, providing a description, summary, and critical evaluation of each work. The purpose is to offer an overview of significant literature published on a topic.

2. Components

Similar to primary research, development of the literature review requires four stages:

- Problem formulation—which topic or field is being examined and what are its component issues?
- Literature search—finding materials relevant to the subject being explored
- Data evaluation—determining which literature makes a significant contribution to the understanding of the topic
- Analysis and interpretation—discussing the findings and conclusions of pertinent literature

Literature reviews should comprise the following elements:

- An overview of the subject, issue or theory under consideration, along with the objectives of the literature review
- Division of works under review into categories (e.g. those in support of a particular position, those against, and those offering alternative theses entirely)
- Explanation of how each work is similar to and how it varies from the others
- Conclusions as to which pieces are best considered in their argument, are most convincing of their opinions, and make the greatest contribution to the understanding and development of their area of research

In assessing each piece, consideration should be given to:

- Provenance—What are the author's credentials? Are the author's arguments supported by evidence (e.g. primary historical material, case studies, narratives, statistics, recent scientific findings)?
- Objectivity—Is the author's perspective even-handed or prejudicial? Is contrary data considered or is certain pertinent information ignored to prove the author's point?
- Persuasiveness—Which of the author's theses are most/least convincing?
- Value—Are the author's arguments and conclusions convincing? Does the work ultimately contribute in any significant way to an understanding of the subject?

3. Definition and Use/Purpose -- A literature review may constitute an essential chapter of a thesis or dissertation, or may be a self-contained review of writings on a subject. The literature review itself, however, does not present new primary scholarship. In either case, its purpose is to:

- Place each work in the context of its contribution to the understanding of the subject under review
- Describe the relationship of each work to the others under consideration
- Identify new ways to interpret, and shed light on any gaps in, previous research
- Resolve conflicts amongst seemingly contradictory previous studies
- Identify areas of prior scholarship to prevent duplication of effort
- Point the way forward for further research

- Place one's original work (in the case of theses or dissertations) in the context of existing literature

EXAMPLE of a literature review in the social sciences – *go to lib.umn.edu, click "E" under "Articles & More", click on "EBSCO MegaFILE" and search for:*

Allen, R.C. (1996). Socioeconomic Conditions and Property Crime: A Comprehensive Review and Test of the Professional Literature. *The American Journal of Economics and Sociology, 55*, 293.

(Retrieved on 5/27/08 from http://library.ucsc.edu/ref/howto/literaturereview.html)

Organizing your Literature Review and your Sources

- Create an outline of your argument and then use it as a framework for the literature search
- Qualitative researchers may choose to organize literature review by:
  - Chronological methods - write about the materials according to publication date or by looking at trends (i.e., history of whaling – dividing each section into eras)
  - Thematic methods - organized around a topic or issue, rather than the progression of time (however, time progression may still be important within the themes)
  - Methodological methods – focusing on the methods of the researcher or writer
- Quantitative researchers may choose to organize literature review by:
  - Review of independent variables
  - Review of dependent variables
  - Review of scholarly literature that relates dependent and independent variables

Prioritizing your Material for a Literature Search

- Journal articles in peer reviewed (juried) journals especially those that describe specific research projects. The research may be quantitative and qualitative depending on the nature of your project.
- Monographs or Books
- Conference Papers/Proceedings- particularly for areas that are new or not well represented in the research literature.
- Using the search engines AND digging in the stacks
- Using the abstracts to decide if a particular article will actually be helpful to you
- Using MAJOR AND MINOR keyword descriptors to find most relevant research

Succinctly Summarizing Sources

For research articles:

- State the problem being addressed
- Central purpose and concept of study
- Sample, population, subjects
- Highlight KEY results
- Point out methodological limitations if any
- Implications for areas of further study

For Essays, Non-empirical Work and/or Research Synthesis:

- Briefly articulate problem being addressed
- Identity primary theme of the piece
- Highlight major conclusions related to theme
- Specify practical application of research outcomes
- Critique methods, logic of argument if necessary

Important Do's and Don'ts

- Do create a cogent, thoughtful argument about the importance of your research
- Do use the literature that gives your ideas a foundation to stand on
- Do include contradictory ideas— but remember to seek out means of countering them.
- Do organize your review so that there is smooth transition between studies cited.
- Do demonstrate the relationships between the works you are citing.
- Do cite everything correctly—be overly cautious if you need to.
- Don't use a study simply because you have it—does it serve a purpose?
- Don't simply summarize one study after another without demonstrating a connection or transition between issues/ findings.
- Don't create an Annotated Bibliography or book report—this is a literature review.
- Don't forget to proof read for grammar and spelling errors.

Adapted by R. Jehangir, 2003, from Creswell, J. W. (1994). *Research Design: Qualitative and Quantitative Approaches*. Thousand Oaks, CA: Sage Publications

Citation Styles … What's Yours?



Do you know these styles and the differences among them?

☑ Review citation styles by going to the following website: http://en.wikipedia.org/wiki/Citation

☑ As you can see, disciplines vary in the style of their citations. What does Wikipedia state the appropriate style of citation for your discipline is?

_____

☑ Now cross-check this with your Faculty Mentor – what do they report to be the appropriate citation style for your research?

_____

☑ List two style guide resources (printed or online manuals) that you will use to properly cite in your discipline's style.
Ex. - APA style http://owl.english.purdue.edu/owl/resource/560/01/

1. _____

2. _____

Week 2

Personal Statement

The personal statement is one of the most important components of a graduate school application. A personal statement allows the admissions committee to gain a better understanding of your academic and professional goals, academic background, research experience, and compatibility with the graduate department/program, among other things. Therefore, aside from making sure your grammar and spelling are correct, it is wise to make sure your personal statement includes the following five parts:

1. Your objective (s)—such as, "My long term goal is to obtain at Ph.D. in political science to research and instruct American Indian law and sovereignty-related courses at a major research university. I also intend to use my training to aid in Indian Child Welfare Act policy reform as it pertains to group and community homes that house American Indian adolescents." In this section you should foreground what you are going to say in the following four sections.

2. Academic experience—here you want to discuss your academic background as it pertains to your objectives. Talk about your degree(s) and how it relates to your current goals.

3. Professional goals—here you want to talk about what you would do if you had that Ph.D. Do not be broad here; be specific. The admissions committee wants to know specifically what you want to do. Do you want to teach, work in the areas of policy and/or the private sector? Make sure what you write here relates to your first section on objectives.

4. Research experience—this is probably the most important aspect of your personal statement. You definitely want to discuss your research experience (McNair and others) and how it relates to your objectives. However, also discuss the research you **would do** if in the graduate department. Also discuss some of the faculty that you would like to work with on that prospective research project. Do not forget to situate your research interests within the graduate department. In other words, relate your proposed research to the graduate department you are applying to.

5. Conclude with a summary of why this program is tailored to your interests intellectually and professionally. Something like, "The Ph.D. program in the Department of Political Science at the University of Minnesota is the best place for me to achieve my intellectual, professional and community endeavors. I have spoken to, and worked with, a number of faculty in your program, such as…"

After you have the five basic parts of your personal statement, do not forget to add such things as: relevant work experiences related to your objectives, hardships or other obstacles you have faced. **Use specific examples or short stories to make it personal!** Write about your life, family, community and anything else that sets you apart from the rest of the applicants.

Your statement should be about two pages, double spaced or 1.5; 10 or 12 pt font. Be sure your personal statement, CV and any other documents you provide are all formatted the same (or very nearly the same) -- and don't forget to put your name on it!

To watch a 6 minute online workshop on writing a personal statement, go to: https://umconnect.umn.edu/personalstatement



Characteristics of a *Good* Personal Statement

Before you write a single word, make sure you know what is expected of a successful graduate school essay.

A good essay…

-Is thoughtful and honest

A strong personal statement is **reflective**; that is, it demonstrates that you have thought about and gained a clear perspective on your experiences and what you want in your future. It does not simply tell a reader what you think he/she wants to know. Instead, it gives the reader a vivid and compelling picture of you--in essence, telling the reader what he or she should know about you. Remember that the focus of the essay is YOU--your achievements, your obstacles, your goals, your values.

-Strives for depth, not breadth

A good essay is not a list of your accomplishments. Remember when your mom told you that it's quality, not quantity that counts? Well, the same adage applies for your college essay. A reader will be much more interested in how your experience demonstrates the theme of your essay, not the number of accomplishments you can list. What is NOT interesting: an essay that devotes one paragraph each to a variety of different topics. This type of approach denies you the ability to give depth to your essay.

-Follows the conventions of good writing

A good essay uses appropriate grammar and syntax, uses precise and vivid language, and does not contain any spelling errors.

-Conforms to guidelines

If the essay instructions tell you that the essay should be two pages long, on white 8.5x11 inch paper, then the essay should be two pages long, on white 8.5x11 inch paper. Less is not more, and more is not better, either.

-Answers the question!

A good essay is the result of a writer who has examined the essay question and written an essay that explicitly addresses that question. For example, if you are asked to *describe* your greatest accomplishment or any unusual circumstances or challenges you have faced, then your reader will expect you to use vivid language that will enable the reader to visualize your accomplishment and share your sense of success.

-Benefits from several drafts and feedback from others

Revision allows an essay to grow. Revising is not editing; revising is the act of "re-seeing" and of looking for those parts of the essay that would benefit from more explication, more (or less) vivid language, or even deleting parts that simply don't work to move your primary theme forward. Similarly, feedback from others can help you identify those parts of the essay that work well--and those that don't.

-Contains a catchy introduction that will keep the reader interested

It is important to recognize that essay readers will read hundreds, maybe even thousands, of essays during the application review period. That means that an essay with a catchy introduction, one that gets right to the point and uses precise language and vivid imagery, is going to stand our more than an essay that is predictable and conventional in its opener.

-Transforms blemishes into positives

It's okay to have flaws! The essay is your chance to show how you have transformed blemishes. For example, if your essay theme is "overcoming obstacles" and you earned a poor grade in a class, but went to a community college at night to repeat the course, it is important for your reader to know this because it is an example of your perseverance. The reader does not want to hear complaints about poor grades or circumstances, but rather wants to know how you have overcome them.

-Demonstrates your knowledge of the major/college

No one expects you to know everything about the college or university to which you are applying. However, readers will want to know that you have done your homework. For example, if you write an essay that states your interest in becoming an engineer, but the college does not have an engineering program, then you haven't done your homework.

-Exudes confidence--you will be successful no matter what

A good essay doesn't beg or brag. Colleges and universities want to admit the best students, and the best students are those who can demonstrate their ability to pursue their goals regardless of where they are admitted. Think of this as quiet

confidence--the kind that reveals itself through your description of lifelong interests, sustained commitment, and/or perseverance in the face of adversity.

(Adapted from http://students.berkeley.edu/apa/personalstatement/gettingstarted.html)

8 Personal Statement Blunders



Just as you should know what to do, you should also know what NOT to do! Here are some of the biggest blunders students make in their essays:

1. The essay repeats information contained elsewhere in the application

Sometimes students, to be on the safe side, simply repeat in the essay the same information that is in the application itself. This strategy results in the reader gaining no more insight into what drives you than he/she discerned from the rest of the application packet. Remember, your reader already knows from your application, for example, that you are in the California Scholarship Federation and a member of the Ethnic Studies Club. What the reader doesn't know is *why* you chose to participate in these activities and *how* your involvement in these activities is evidence of your particular interests and talents--your essay's theme. If one of these experiences is a good example of your essay's theme, then by all means include it. If you're just including it because you think that you'll impress the reader with everything you've ever done, think again.

Example:

In my junior year I was a cheerleader for my school. I worked really hard at it, and found it to be fun and challenging. I was also part of my school's Kids in the Kitchen program, which helped to make food available to poor people in my community. Cheerleading and volunteer work kept me very busy. I spent approximately twenty hours each week cheering and another five hours volunteering. I learned a lot from this experience and can manage my time effectively and maintain a positive attitude in the face of adversity.

2. *The writer complains about his/her circumstances rather than explains them.*

Remember that admissions officers want to know how resilient you are. While it is certainly okay to write about obstacles you've faced, what is important to your reader is how you overcame the obstacle, not what a terrible obstacle it was.

Example:

Because my mother is a single parent, she has had to make a lot of sacrifices to keep me and my brother in a private school. It means that we have to go without a lot of things, which is sometimes embarrassing. But even though everyone in my school knows that we are poor, no one is willing to give me a break. This is

especially true of my English teacher, Sister Magdalena. Because she didn't like me, and she is not comfortable with poor people, she gave me a C in English when I really should have gotten a B.

. ▪ . ▪ . ▪ . ▪ . ▪ . ▪ . ▪ .

3. *The writer discusses money or a college's ranking as a motivating factor for applying to a particular major/college.*

Yes, we all want to attend college to earn more money. And we all want to attend the most prestigious colleges. But college faculty who read your essay want to know that you are motivated by a love of learning. So, even though money or a school's ranking may be important to you, keep this information out of your essay.

Example:

I want to study engineering because a recent US News and World Report article said that engineering is the fastest-growing industry in the nation and the best place to study engineering is UCLA. With a degree in engineering, I will be able to buy a house for my mom.

. ▪ . ▪ . ▪ . ▪ . ▪ . ▪ . ▪ .

4. The essay relies on gimmicks rather than substance.

A "gimmicky" essay is one in which the reader tries to get the reader's attention through unconventional means. This does not mean that your essay has to follow one set format; what it means is that gimmicks can't replace substance.

Example:

College, oh college/How much I want thee/for college, oh college/will strengthen me/and with a degree in hand/I will change this land/and make a better life for you and me. (This is supposed to be a poem.)

. ▪ . ▪ . ▪ . ▪ . ▪ . ▪ . ▪ .

5. The writer makes claims in the essay that are not backed up by the application.

The essay is a component of the application and is read within the context of the application. A description of yourself as the top student in the school should be supported by your grades. Similarly, claims made about your extracurricular experiences should be backed up by the application. For example, a student who claims that her lifelong ambition is to save the environment would want this claim supported by examples of involvement in environment-related hobbies, clubs and classes.

6. *The essay contains the wrong school name*

Oops! In these days of computerized cutting and pasting, this is an easy blunder to make. Proofread carefully!

7. *The essay contains mechanical errors or errors of usage, clichés, or meaningless prose*

Although your reader is not grading your essay or scrutinizing your grammar, a poorly written essay signals a reader that you are unfamiliar with conventions of good writing or simply did not put enough time into composing your essay. Either way, there will be other applications whose essays are very polished, so don't disadvantage yourself.

Examples:

My father always told me that there is nothing to fear but fear itself. (A cliché)

A feeling of indescribable disbelief overcame me. (Wordy prose)

8. *The essay is too long or too short*

Show that you know how to follow directions. An essay that is too short may indicate carelessness; one that is too long may signal arrogance. Remember that your readers have many other applications to read, so be sure that the limited time available to peruse yours will be spent reading an essay that is the appropriate length.

(Adapted from http://students.berkeley.edu/apa/personalstatement/blunders.html)

Examples of

Great Personal Statements !

Discipline – Education

As a single mother who has confronted homelessness and poverty, I am committed to reforming public education. I have a particular interest in improving the accessibility of schooling to homeless individuals, single mothers, and disadvantaged women of color. As a result of my own experiences, I am familiar with the despair and frustration endemic to individuals struggling to survive. I am convinced that increasing individuals' access to education can make the difference between despondency and hope. I am applying for my doctorate in educational leadership so that I can pursue my life-mission: to make education accessible to all.

I began preparing for this mission by volunteering as an intake coordinator at the Christian Assistance Ministry (CAM). Although I had many responsibilities, the role that consumed most of my time and energy was interviewing clients and assessing their physical and emotional needs. One of my greatest contributions was streamlining and updating the paperwork associated with CAM's intake process. I also generated an updated resource list that included many service agencies in the Houston area. By personally contacting each agency to acquire contact information and to learn about its services, I facilitated greater communication between service organizations and ensured that my clients had access to necessary aid.

In my present position as Research Analyst at SeaNet, my primary role is ascertaining the needs of client networks. As an umbrella group, SeaNet has only limited contact with small business development centers, and my job is to ensure that our organization meets these centers' needs. When I took the initiative to send out surveys asking agencies to rate our group's effectiveness, I received an overwhelming response. The information I compiled was so revealing that it was published in the quarterly report that is sent to our funding agency in Washington, D.C. In addition, I have been involved in a number of special, innovative projects. Recently, I analyzed the availability of renewable energy resources in Southern Texas in tandem with The Economic Development Center, Solar Energy, Brooks Air Force Base, and research universities in Texas.

As part of my master's thesis, I collaborated with Upward Bound, Peace Center, and the Davis Education Foundation to underline the importance of community cooperation in public schools. I also provided informational brochures and handouts detailing other such organizations that could assist with the individualized needs of schools. I was gratified when my efforts resulted in teachers and administrators contacting several of the organizations I had mentioned so that the organizations could start outreach in their educational districts.

Although I have not yet been employed in the educational sector, my master's work, as well as my life experiences, has given me a nuanced and sophisticated knowledge of the educational field. I have acted as a mentor at Davis Middle School for many years and have tutored a number of home-schooled children. When my own children attended school, I was involved in their schools' organizations and often took on a leadership role on educational committees. I served on many boards and was active in assisting both instructors and administrators. Fifteen years of experience has familiarized me with the diverse needs of Houston's students, and it has prepared me to act on their behalf.

My short-term goals include advancing my knowledge of quantitative research using programs such as SPSS and Microcas, and acquiring a sophisticated understanding of how to become a leader in the educational field. I wish to use these skills to promote empirical studies in education that can help direct educational reform.

I am attracted to the doctoral program at Texas A&M for precisely this reason. Only Texas A&M offers the kind of collaborative experience that suits my personal needs and professional goals. Over the past four years, I have seen the department blossom into a challenging and innovative program. I am

excited about the program's direction and its emphasis on practical application. I appreciate that every course offers the opportunity for independent research, and that the faculty is open to student suggestions for improvement. The flexibility and patience of the faculty and the talent of the students makes Texas A&M a singular choice for my doctoral degree.

My long-term goal is to use my past experience and Texas A&M training to help make education accessible to all, particularly the underprivileged. Whether I am working in the public or private sector, I look forward to addressing the nation's urgent need to educate its citizens efficiently and comprehensively.

Discipline - Chemistry

"This semester," bellowed the teacher as he addressed the room of high school seniors, "I will make you all chemists." The class let out a collective groan. The teacher continued unfazed. "You will all have to complete a real research project using the facilities of local universities." Like most of my classmates in the AP Chemistry course, I dreaded the huge amount of work that loomed before me. Little did I know that this work would help determine my future. In the months ahead, I committed more time to this one course than I ever had before, poring over chemistry textbooks and working in the laboratory into the early hours of the morning. I always had an inkling that my future lay in science and technology, and this experience proved it beyond question. The methodical laboratory preparations, the interweaving of theory and practice, and the tangible results of a successful experiment all appealed to my disciplined and inquisitive nature. By the end of my senior year, I had developed a solid foundation in the principles of chemistry, and just as my teacher had promised, I was on my way to becoming a chemist.

Despite my growing interest in chemistry, I still harbored dreams of becoming a fighter pilot in the United States Air Force at the time of my high school graduation. Even as I spent my evenings in the laboratory for AP Chemistry, I woke up early in the morning to run and do a routine of sit-ups and pushups. I was preparing for the rigorous physical and mental demands of the Air Force Academy. Over the course of previous summers, I had experienced the sweeping rush of glider school and had soloed in my first motorized aircraft. I hoped for the challenging life of an Air Force pilot and was on my way to making it a reality through sweaty self-discipline and intense mental focus. On one sunny afternoon, I saw a thick letter with a Colorado postmark in my mailbox. I was now a cadet at the Academy.

As with all military institutions, the first years at the Academy were intensely demanding. Strenuous endurance tests, high-level academic work, and hazing by upperclassmen made a junior cadet's life extremely trying. Nonetheless, I thrived during these years because I never lost my sense of focus or my love of chemistry. Even as I dreamed of being a fighter pilot, I found my chemistry courses to be highly engaging and challenging. All cadets were given a choice of academic tracks, and I had chosen science and engineering. As in high school, I worked in the laboratory late at night and ran early in the morning. Whenever I got fatigued, I just remembered how close I was to fulfilling my goal of becoming a pilot.

After a visit to the ophthalmologist one day, my dreams of airborne glory were abruptly shot down. My vision did not meet the Air Force standards required of pilots. It was a crushing blow, and at first I did not know what to do. Nonetheless, I knew that I was not without direction. Once again, my life pushed me towards studying chemistry. This was a subject that I knew I would enjoy, so I packed up my science textbooks, threw away my flight training manuals, and headed back to my hometown college, the University of Texas at Austin (UTA).

As much as I had loved chemistry before, my studies at UTA convinced me that I had found the right vocation. I was particularly inspired by Dr. John Rove. Like my high school chemistry teacher, Dr. Rove knew that the best way to instill an appreciation for the field was through active research and meaningful laboratory work. He believed that one should never learn from a book what one could learn from a beaker, flask, or Bunsen burner. As soon as I explained my experience and interest in the field

to him, he offered me a position in his group doing research on the decomposition mechanisms of electro-optic dyes. The research I performed involved finding suitable organic dyes for electro-optic modulators. After months of grueling work and wise guidance from Dr. Rove, I was ready to present my research at regional and national meetings of the American Chemical Society. Networking with chemists from around the country allowed me to glimpse not only other possible fields of chemical research, but also the potential career options I might want to pursue. Had it not been for Dr. Rove's personal guidance and help, I doubt that my future career plans in chemistry would be as clear as they are today.

Because of my two-year commitment to the United States Air Force, I was unable to apply to graduate school directly after college. Instead, I spent two years serving as an intelligence officer. From this job, I have greatly improved my analytic ability, communication skills, pressure management techniques, and patience as a researcher. Like chemists, intelligence officers often have to construct coherent models of reality from incomplete pieces of data. This position has offered me insights and training that no classroom setting could match. Nonetheless, I am ready to return to my first love, chemistry.

Although my life has taken some unexpected turns in the last few years, the field of chemistry has been an unwavering companion and a source of intellectual interest. Whether it was discovering research in high school, learning advanced chemical theories at the Air Force Academy, or presenting my findings at meetings of the Chemical Society, the lure of studying physical structures has always guided my decisions. My advanced research in electro-optic modulators and my course work in classes such as Molecular Orbital Theory have given me the analytical ability, laboratory skills, and theoretical knowledge to excel in a rigorous doctoral program. During my Ph.D. studies, I intend to focus on inorganic and materials chemistry in order to design new materials for use in industry. By a stroke of fate, the weak vision that prevented me from becoming a fighter pilot allowed me to envision my future as a successful chemist.

(Retrieved 5/27/08 from http://www.essayedge.com/promo/samplework.shtml)

Week 3

Methodology

QUANTITATIVE RESEARCH DESIGNS

Subjects
Describe the subjects. Focus here on sample size, and the proportion in the sample of males, females, various ethnic groups, and age.

Design
Describe the overall research plan, and type of research, i.e. experimental, survey, case study.

Variables
Describe the variables used, characteristics of measurements, reliability and validity. Here, go into detail about your *independent* variable; that is, the variable that you are manipulating. Also, go into detail about your *dependent* variable; that is, the variable you are measuring. If you are using surveys or other instruments, describe these instruments. How were they normed? What are their validity and reliability coefficients?

Use operation definitions to discuss your variables. For example, how is self-esteem defined in your study? What instrument did you use to operationalize this construct? *Operational definitions* are important. These definitions allow anyone interested to reproduce your study using exactly the same measures.

Procedures
Describe the steps required to carry out the data collection, especially what happens to subjects if you have them.

Statistical Hypothesis
There are two types of hypotheses. The research hypotheses which is what you are expecting to find, and the null hypotheses which is the hypothesis of no difference or the status quo. The hypothesis you test is the null hypothesis.

Analysis
Here you want to name the specific statistical procedure you used to analyze your data.

QUALITATIVE RESEARCH METHODS

Type of Design

- Tell the reader specifically about the design of the study— the *approach* to data collection, analysis and report writing (go back to table of Research Paradigms)
- Describe the characteristics of design
    - Typical unit of measure
    - Any other specific characteristics of design
    - Subjects or informants in the study (may need to be especially conscious of anonymity here)

Researcher's Role

- In some studies it is appropriate to include statements about the researcher's past experiences with the topic, the setting or the subjects
- Steps taken to gain access to the subjects – if Institutional Review Board Permission was required indicate it here
- Comment on any sensitive ethical issues that need to be taken into consideration to maintain anonymity of subjects

Data Collection Procedures

- Indicate types of data collection used
    - Interview
    - Observations
    - Focus Groups
    - Document Analysis (Meeting Minutes, Newspapers)
    - Audio Visual (photographs, art, film)

- Qualitative data analysis procedure can be eclectic—unlike quantitative procedures which make distinct separations between procedures qualitative procedures can simultaneously engage in numerous procedures
    - Collecting from the field, sorting into categories, formatting the information
- Describe the process of "reduction" or interpretation that you will use to organize the data
- Discuss matrices or coding systems/ programs you will be using

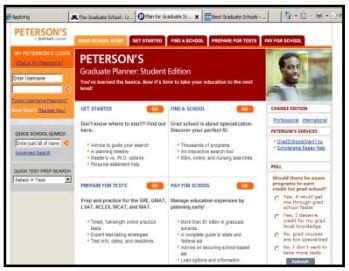
Verification

- Discuss the issues of internal validity—accuracy of the information and whether it matches reality
    - Member checks
    - Inter-rater procedures
    - Triangulation
- Discuss limitations to generalizability
    - Remember that the purpose of qualitative data is not to generalize findings but to form an interpretation of events but you still need to indicate limitations to generalizability

Week 4

Finding & Evaluating Graduate School Programs

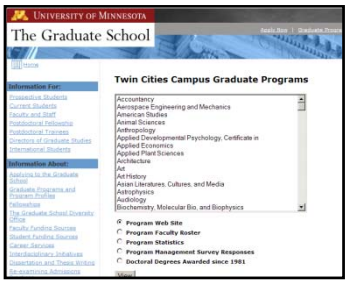
ONLINE RESOURCES.



Graduate School Information
GradSchools.com http://www.gradschools.com/

Peterson's http://www.petersons.com/

Rankings Websites
US News – America's Best Graduate Schools
http://grad-schools.usnews.rankingsandreviews.com/grad



University Websites (examples)
Stanford University http://gradadmissions.stanford.edu/

University of Minnesota http://www.grad.umn.edu/

Diversity Resources (always check for this on university websites)
University of Minnesota http://www.grad.umn.edu/outreach/

Schools that Support McNair Scholars
Application fee waivers:
http://www.rochester.edu/college/kearnscenter/mcnairgradfeewaivers.html
Fellowships: http://www.rochester.edu/college/kearnscenter/mcnairgradfellowships.html

TALK TO PEOPLE. Ask professors, advisors, current grad students, or alumni of graduate programs for

- Their recommendations for well-regarded graduate programs in your field
- Recommendations of programs that fit your individual needs (location, funding, faculty who study what you want to study, etc.)
- Their experience in graduate school and what was most important for them in finding a good "fit" between themselves and their graduate program
- What they wish they would have known about searching for grad schools

TIPS.

- Spend time researching individual programs
- Request application materials and program handbooks
- Go to graduate school fairs (but never pay to go to a fair)
- If possible, visit the school (sometimes schools will pay for your visit, especially through diversity recruitment /outreach efforts)
- Find out about specific courses offered
- Make contact with current students. What is the program's emphasis?
- Find out how long it takes to complete the program, and inquire about the program's attrition rate (how many students left the program, and why did they leave?)
- Learn about financial aid resources available from the school and from the program. Are Teaching or Research Assistantships available to incoming students? What types of fellowships (scholarships) or grants are available? Do you qualify for loans?
- Learn about specific research interests of individual faculty members

WHAT DO I DO WITH ALL OF THIS INFORMATION?
Create a list of potential schools and programs. Ideally, this list may include ~10 programs. Past McNair Scholars have found it helpful to organize and rank their lists based on their potential for being accepted (**Safety, Good Chance, Reach**). *Don't discard schools simply because you feel your chances of acceptance are slim!*

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Graduate School Choices

Create a list of potential schools and programs. Organize and rank your list based on your potential for being accepted by picking 3 schools/programs that meet each of the following categories:

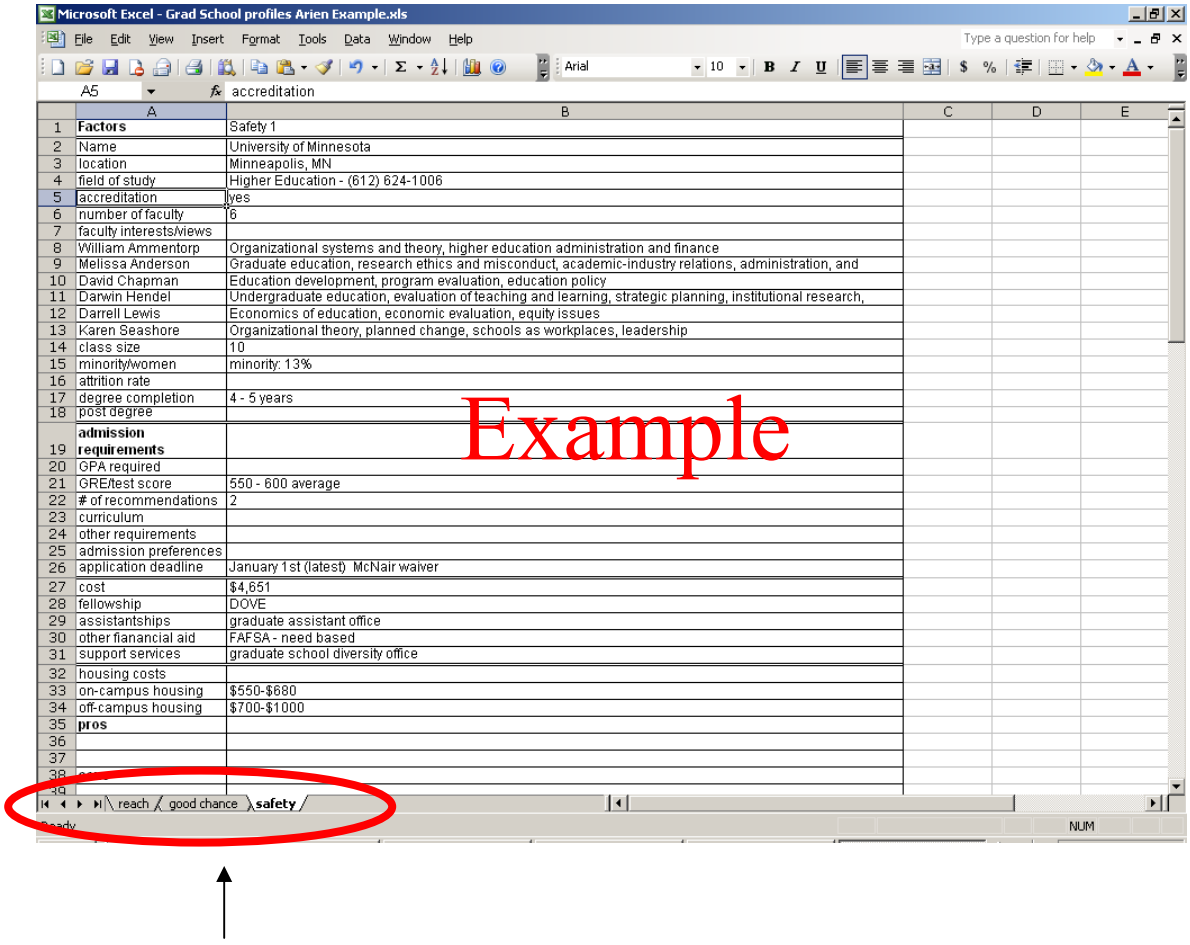
SAFETY: Meet all the requirements; they might have recruited you, or you have good relationships with professors

GOOD CHANCE: Meet some or almost all of requirements

REACH: Never would apply to it, do not meet many of the requirements, yeah right!

Use the electronic spreadsheet (we'll email it to you) or fill out the forms on the following pages.

Example of completed electronic spreadsheet:



	A	B
1	**Factors**	Safety 1
2	Name	University of Minnesota
3	location	Minneapolis, MN
4	field of study	Higher Education - (612) 624-1006
5	accreditation	yes
6	number of faculty	6
7	faculty interests/views	
8	William Ammentorp	Organizational systems and theory, higher education administration and finance
9	Melissa Anderson	Graduate education, research ethics and misconduct, academic-industry relations, administration, and
10	David Chapman	Education development, program evaluation, education policy
11	Darwin Hendel	Undergraduate education, evaluation of teaching and learning, strategic planning, institutional research,
12	Darrell Lewis	Economics of education, economic evaluation, equity issues
13	Karen Seashore	Organizational theory, planned change, schools as workplaces, leadership
14	class size	10
15	minority/women	minority: 13%
16	attrition rate	
17	degree completion	4 - 5 years
18	post degree	
19	**admission requirements**	
20	GPA required	
21	GRE/test score	550 - 600 average
22	# of recommendations	2
23	curriculum	
24	other requirements	
25	admission preferences	
26	application deadline	January 1st (latest) McNair waiver
27	cost	$4,651
28	fellowship	DOVE
29	assistantships	graduate assistant office
30	other fianancial aid	FAFSA - need based
31	support services	graduate school diversity office
32	housing costs	
33	on-campus housing	$550-$680
34	off-campus housing	$700-$1000
35	**pros**	
36		
37		

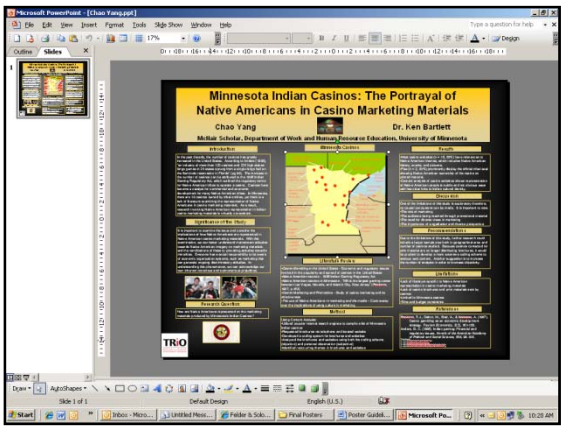
Example

Please note that there are three tabs within this spreadsheet: one for Reach, one for Good Chance, and one for Safety.

Week 5

McNair Scholars
Poster Guidelines

Create your research poster using one very large slide in PowerPoint. →



Poster Size: 40'' wide and 32" high (specify size in PowerPoint 2003 by going to File→Page Setup→Custom)

Title: Be as accurate as possible using 100-120 pt. Arial font that is **bold** and CAPITALIZED.

Author Names: 36-42 pt. Times font, **bold** – list faculty name, department, your name. Check to see that spelling and department names are correct.

Section Headings & Titles: 36 point Times font, **bold** – include the sections your mentor feels are most important to communicate your research. This typically includes an introduction or abstract, a hypothesis or research question, methods, results, discussion, limitations, and/or key references. However, this will vary based on your discipline and research project. Indicate spaces for any charts/graphs/diagrams you plan to use.

Text: 28 pt. Times font

Margins: 1.5"+ on all sides

Background Color: Use a light background with a dark font or a dark background with a light font. The printing will show up better when there is a strong contrast between background and the typing.

Image Resolution: If you plan to include photos, images or graphs, the image resolution is critical. Be careful of using web images, as many are copyrighted and may have poor resolution once blown up onto a poster.

Include TRiO logo: This will be emailed to you.

Viewing Tip: To see the whole poster at once, view at ~17%. To view and edit sections of your poster, view at ~66%. To see the how the poster actually looks, view at 100% (this is especially important, because you will see how the boxes/text/graphics actually line up – any view other than 100% will be slightly skewed).

Printing Tip: Use print preview, then "Scale to Fit" to view and print the poster in a one page 8.5"x11" format. When we print the final version of the poster, we will use a special large-format printer in Coffman.

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Examples of Past Posters



Minnesota Indian Casinos: The Portrayal of Native Americans in Casino Marketing Materials

Chao Yang — Dr. Ken Bartlett

McNair Scholar, Department of Work and Human Resource Education, University of Minnesota

Introduction

In the past decade, the number of casinos has greatly increased in the United States. According to Anders (1998), "an industry of more than 120 casinos and 220 high stakes bingo games in 24 states sprung from a single bingo hall on the Seminole reservation in Florida" (pg 59). The increase in the number of casinos can be attributed to the 1998 Indian Gaming Regulatory Act, which outlined the regulatory terms for Native American tribes to operate a casino. Casinos have become a catalyst for commercial and economic development for many Native American tribes. In Minnesota, there are 19 casinos owned by tribal entities, yet there is a lack of literature examining the representation of Native Americans in casino marketing materials. As a result, research involving Native American representation in Indian casino marketing materials is virtually nonexistent.

Significance of the Study

It is important to examine the issue and consider the implications of how Native Americans are represented in Native American casino marketing materials. With this examination, we can better understand mainstream attitudes towards Native American imagery on marketing materials and the ramifications of these to prevailing attitudes towards minorities. Everyone has a social responsibility to be aware of economic organization systems, such as marketing that can generate ongoing discriminatory attitudes. In understanding this phenomenon, we can acknowledge our own inherent conscious and subconscious prejudices.

Research Question

How are Native Americans represented on the marketing materials produced by Minnesota's Indian Casinos?

Minnesota Casinos

Literature Review

- *Casino Gambling in the United States* - Economic and regulatory issues involved in the popularity and spread of casinos in the United States
- *Native American casinos* - 1988 Indian Gaming Regulatory Act
- *Native American casinos in Minnesota* - "MN is the largest gaming center between Las Vegas, Nevada, and Atlantic City, New Jersey" (Rephann, 1997, p.183)
- *Casino Marketing and Promotions* - Study of casino marketing and its effectiveness
- *The use of Native Americans in marketing and the media* – Controversy over the implications of using culture in marketing

Method

Using Content Analysis:
- Utilized popular internet search engines to compile a list of Minnesota Indian casinos
- Requested brochures via telephone and located website
- Developed a coding system for brochures and websites
- Analyzed the brochures and websites using both the coding scheme (objective) and personal observation (subjective)
- Identified reoccurring themes in brochures and websites

Results

- Most casino websites (n = 15, 83%) have references to Native American themes, which includes Native American history, events, and pictures.
- Few (n = 2, 33%) prominently display the official tribal seal showing Native American ownership of the casino on printed material.
- Content analysis of casino websites shows representation of Native American people is subtle and not obvious ways with few clear links to Indian cultural identity.

Discussion

One of the limitations of this study is exploratory therefore; no causal conclusions can be made. It is important to note:
- The role of marketing
- The audience being reached through promotional material
- The need for diverse views in marketing
- The importance of a egalitarian and diverse perspective

Recommendations

Due to the limitations of this study, further research could include a larger sample size both in geographical area and number of casinos studied. Because casinos contacted for print material are no longer distributing brochures, it would be prudent to develop a more extensive coding scheme to analyze web content. Another suggestion is to increase the number of analysts in order to increase objectivity.

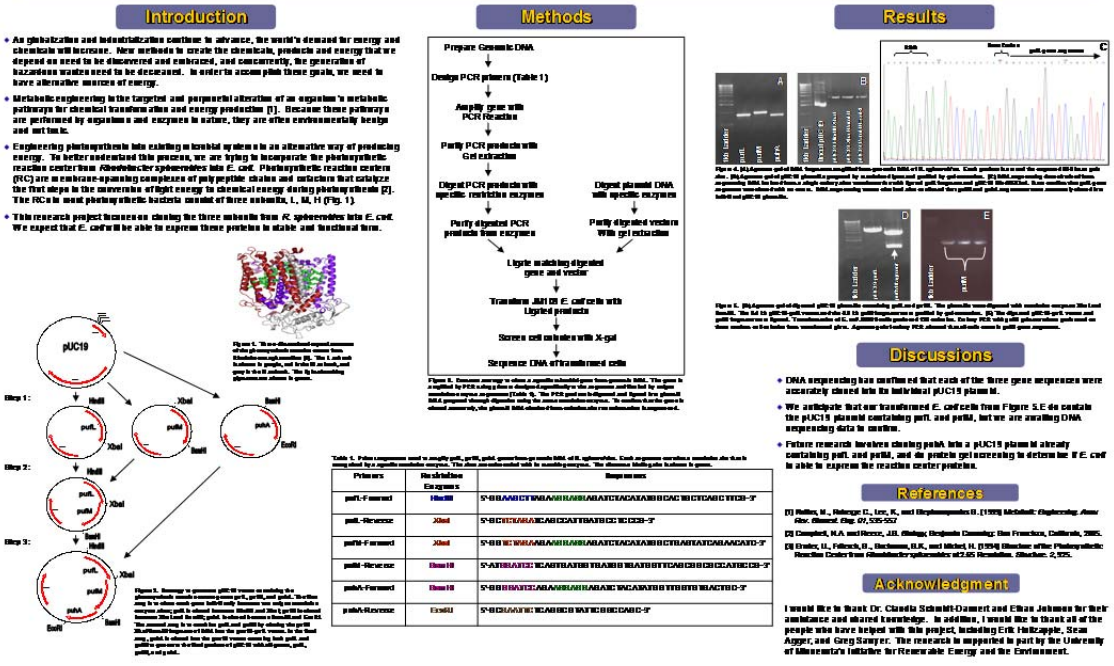
Limitations

- Lack of literature specific to Native American representation in casino marketing materials
- Lack of casino brochures and print materials sent by casinos
- Limited to Minnesota casinos
- Time and budget constraints

References

Rephann, T.J., Dalton, M., Stair, A., & Isserman, A. (1997). Casino gambling as an economic development strategy. *Tourism Economics*, 3(2), 161-183.

Anders, G.C. (1998). Indian gaming: Financial and regulatory issues. *Annals of the American Academy of Political and Social Science*, 556, 98-108.



Engineering *Escherichia coli* to Express Functional Photosynthetic Reaction Center Proteins from *Rhodobacter sphaeroides*

Rae Lee[1], Ethan Johnson[2], Claudia Schmidt-Dannert[2]

[1]Department of Biology, Saint Olaf College, Northfield, Minnesota

[2]Biochemistry, Molecular Biology and Biophysics, University of Minnesota Twin Cities, Saint Paul, Minnesota

Introduction

- As globalization and industrialization continue to advance, the world's demand for energy and chemicals will increase. New methods to create the chemicals, products and energy that we depend on need to be discovered and embraced, and concurrently, the generation of hazardous wastes need to be decreased. In order to accomplish these goals, we need to have alternative sources of energy.
- Metabolic engineering is the targeted and purposeful alteration of an organism's metabolic pathways for chemical transformation and energy production [1]. Because these pathways are performed by organisms and enzymes in nature, they are often environmentally benign and nontoxic.
- Engineering photosynthesis into existing or toxicant systems is an alternative way of producing energy. To better understand this process, we are trying to incorporate the photosynthetic reaction center from *Rhodobacter sphaeroides* into *E. coli*. Photosynthetic reaction centers (RC) are membrane-spanning complexes of polypeptide chains and cofactors that catalyze the first steps in the conversion of light energy to chemical energy during photosynthesis [2]. The RCs in most photosynthetic bacteria consist of three subunits, L, M, H (Fig. 1).
- This research project focuses on cloning the three subunits from *R. sphaeroides* into *E. coli*. We expect that *E. coli* will be able to express these proteins in stable and functional form.

Methods

Primers	Restriction Enzymes	Sequences
pufL-Forward	HindIII	5'-GGAAGCTTGNNAANRRANRANRAAATCTACATANRRGCAGTRGTCGAGCTTGG-3'
pufL-Reverse	XbaI	5'-GGTCTAGANCAGCCATTGATRGCCGCCG-3'
pufM-Forward	XbaI	5'-GGTCTAGANNAANRRANRANRAAATCTACATANRRGCTTAATATCANAACATG-3'
pufM-Reverse	EcoRI	5'-ATGAATTCNCATRATRGTRAGGGTRATGGTTCAGGGCGCCATGCGG-3'
pufA-Forward	EcoRI	5'-GGAATTCNGAANRRANRAGCNCAYATGTTTGTTGACTGG-3'
pufA-Reverse	EcoRI	5'-GGAATTCNCARGCGTATCGGCCATG-3'

Results

Discussions

- DNA sequencing has confirmed that each of the three gene sequences were accurately cloned into its individual pUC19 plasmid.
- We anticipate that our transformed *E. coli* cells from Figure 5.E do contain the pUC19 plasmid containing pufL and pufM, but we are awaiting DNA sequencing data to confirm.
- Future research involves cloning pufA into a pUC19 plasmid already containing pufL and pufM, and do protein gel screening to determine if *E. coli* is able to express the reaction center proteins.

References

[1] Keller, M., Belenger, C., Lee, K., and Stephanopoulos G. (1999) *Metabolic Engineering. Ann. Rev. Biomed. Eng.* 01, 535-557.

[2] Campbell, N.A and Reece, J.B. (Biology) Benjamin Cummings: San Francisco, California, 2005.

[3] Ermler, U., Fritzsch, G., Buchanan, S.K., and Michel, H. (1994) Structure of the Photosynthetic Reaction Center from *Rhodobacter sphaeroides* at 2.65 Resolution. *Structure*. 2, 925.

Acknowledgment

I would like to thank Dr. Claudia Schmidt-Dannert and Ethan Johnson for their assistance and shared knowledge. In addition, I would like to thank all of the people who have helped with this project, including Erik Hultzapple, Sean Agger, and Greg Sawyer. The research is supported in part by the University of Minnesota's Initiative for Renewable Energy and the Environment.

Week 6

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Finishing Up Your Research Paper …

Discussion & Implications

- Specify the most important outcomes/ results of the study (summarize briefly, don't repeat the results section!).
- Based on your results make a clear statement about support or nonsupport of your hypothesis.
- Note how your results were similar or different from other studies in this area.
- Talk about it—what are some plausible explanations for these results
- Specify why you chose to interpret results in the way you did.
- Discuss any limitations of your research study
    - Small sample/ pilot
    - Methodological considerations
    - Any other problems/ confounding variables not accounted for?

Implications are questions derived from your study that merit further exploration.

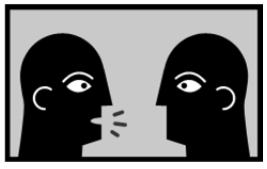
- What has this study contributed to the field?
- How has the study helped answer the hypothesis?
- Conclusions or theoretical implications that can be drawn from this study?

Some additional questions to consider for this section

Have you identified any trends—positive or negative—in the research on your topic? How might negative trends be corrected or positive trends carried further?

Have you noticed any gaps in the research or areas where focus is lacking? Is this a problem, and what should be done?

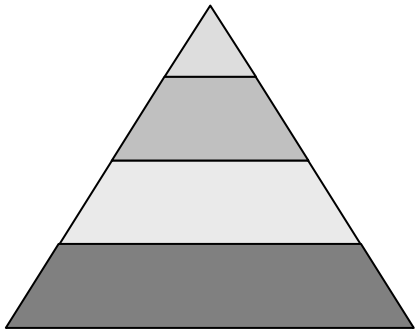
Overall, what direction should your project or discipline go in the future?



Continuing Your Research Paper …

Results

The purpose of a results section is to present and illustrate your findings. Make this section a completely objective report of the results, and save all interpretation for the discussion.



Content
Summarize data collected.
Simple statement of findings.
No discussion about why; be totally objective.
Was hypothesis supported or not?
State if statistically significant.

Do not worry if results seem short.
Results should present actual data, what you found, just the facts!
Results should not include interpretation. Interpretation belongs only in the discussion section.

Results (by type of research)

Historical—use statements from archival/historic documents

Ethnographic—use statements from interviews, quotes from your subjects, observations

Experimental—use descriptive statistics

Figures and Tables

Summarize your findings in text and illustrate them, if appropriate, with figures and tables. Text should complement any figures or tables, not repeat the same information. Do not present the same data more than once. Do not use for data easily presented in a few sentences.

Each figure must be numbered consecutively and complete with caption. Captions go under figures. Each table must be titles, numbered consecutively and complete with heading. Title with description goes above the table. Each figure and table must be sufficiently complete that it could stand on its own, separate from text.

The Final Piece of Your Research Paper!

but it gets placed at the beginning)

Abstract

The abstract is a summary or condensed version of your research paper. It should be one paragraph not exceeding one page in 18 pt. font. Your abstract should be 100-150 words for this research project.

Your abstract should be (stylistic considerations):

- ♦Complete—it covers major parts of the project
- ♦Concise—it contains no excess wordiness
- ♦Clear—it is readable, well organized
- ♦Cohesive—it flows smoothly between the parts

Your abstract should include:

- ♠purpose of research topic/why do it?
- ♠methods/what did you do/how?
- ♠results/what did you find?
- ♠conclusions/logically conclude analysis of data
- ♠relevance/findings relate to literature in field
- ♠describe variables (IV & DV), subjects
- ♠use past tense/since research already conducted
- ♠avoid using bibliographic unless essential

A good abstract shows your research accomplishments and helps you envision your next area of inquiry.

♥McNair staff will help.

Also work closely with mentor to generate successful abstract which reflects essence of research and style in your discipline.

Writing a Smashing Abstract Takes Practice!

Barry Costa-Pierce
University of California
Irvine, CA

An Abstract is the most important part of a scientific paper. It not only summarizes the salient aspects of the paper but also lures a reader into reading it!

It is likely the majority of people who read your paper would only read the Abstract. If you want to attract more people to read your article, it is the Abstract that will do it.

The biggest mistake in writing an Abstract is to mention that such and such "will be discussed". The Abstract is not a place for waffling; rather it is a succinct summary of the exact details of your findings. The most important data and findings are contained in it, NOT left out.

Start out the Abstract by telling exactly what you did and how you did it. Focus on the rationale and ideas of the study and why it's important in the first two sentences. For example, "This study determined impacts on fish community structure by manipulating prey numbers in a tide pool in Oregon, then used Stella simulation software to model community dynamics".

In the next few lines, focus on the materials and methods, and the data generated from the study. Tell the reader how the data were collected, compiled, and state statistical significance(s). Mention any new tools developed.

Avoid using bibliographic references in the Abstract unless they are absolutely essential to understanding the scholarship or results of the study. However, if your article follows directly from a published work and is a major advance on that specific piece of work, do cite the paper in the Abstract.

The end of the Abstract is just as important as the beginning. This is where you want to hook the reader into examining into your paper! In addition, the concluding lines of the Abstract should lead into the first paragraph of the introduction without repeating what has been said. State the implications of your studies to the field of scholarship in which you are working.

Most of us wait until we finish writing the paper to write an Abstract. However, I suggest you make notes as you write the paper about the important parts of each section. When I'm done writing the paper, I read the paper over a few times, making a list of keywords as I read, then organize these (and my thoughts), and draft the Abstract.

With practice, writing good Abstracts will evolve from being a burden to a joy. You will see succinct summaries of your accomplishments appear. These will help you envision your next area of inquiry. In the process you will attract many people to your work; develop lifelong relationships; and contribute to the wonder of science.

http://www.dbc.uci.edu/~sustain/Abstract.html

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McNair Scholars Printing Contract

Please attach a printed version of your poster to this contract. Your work must be completed and your contract singed one hour prior to your arranged printing time. McNair advisors must be given at least one hour lead time for a final review.

Student's Name: _____ _____
First Last

Poster Printing Time: _____

Page Setup: _____

Mentor's Name: _____

Mentor's Department: _____

Mentor's Approval: _____
Signature or e-mail attached with final approval

McNair Advisor Signature: _____

Peer Reviewer Signature: _____

I understand that McNair Scholars will only pay for one poster printing. If I choose to reprint my poster I will pay for it myself. My poster is complete and satisfactory to me, my mentor, and my McNair advisor…

Signed,
_____ date_____

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McNair Scholars Abstract Book Assignment

Description: This year the McNair program will create a full color online 2012 McNair Abstract book. If funding allows, we plan to provide a hardcopy of the book to all scholars. The book will be a compilation of scholar research abstracts, faculty and scholar profiles, and pictures from the summer. Abstract books will be seen by U of M administrators, local, state, and federal representatives, and grad school recruiters. The McNair book will be a highly visible snapshot of the 2012 McNair Scholars program and your efforts. It is paramount this book is uniform in content and of the highest quality. Below is the list of the McNair Scholars Abstract Book publication guidelines. Please read this document carefully and follow ALL directions. The end product will be a one page document that includes five components:

1. Project Title
2. Abstract
3. Faculty Profile
4. Student Profile
5. Dream statement

This assignment is due by **Wednesday, July 25**, in two formats:

1. Hardcopy signed by your faculty mentor and submitted to your McNair advisor.
2. As a word document attachment emailed to your McNair advisor.

   For uniformity, save your document using this format:
   Last name2012finalabstract (e.g. Sawyer2012finalabstract)

Document Settings: Margins: 1 inch **Spacing:** single **Font:** Times New Roman **Font size:** 12 for text body 14 and bold for Headers/Titles

Additional Guidelines by Section (see p. 2 for example of final product. You can delete text and use as a template.)

Project Title
Name
Abstract—150 words
Faculty Profile—100 words---bold faculty mentor title/name at beginning
Scholar Profile—50 words---bold your name at beginning of profile
Dream Statement---Ronald E. McNair stated: "Whether or not you reach your goals in life depends entirely on how well you prepare for them and how badly you want them. You're eagles! Stretch your wings and fly to the sky." "Before you can make a dream come true, you must first have one." In no more than two sentences share what your dream is.

Empowering Youth Engaged in Self-Photography E.Y.E.S PROJECT

Jessica A. Sawyer

Abstract

Discrimination based on gender, ethnicity, and Disadvantaged Business Enterprise (DBE) participation was examined using the prequalification of construction contracts awarded by the New Jersey Transit Authority (NJTA). The questions of Disadvantaged Business Enterprises being less likely to be pre-qualified at high contract dollars and pre-qualified DBEs being less likely to receive contracts by the NJTA were assessed. Businesses certified as DBEs are those which are mainly for people of color, women, and disadvantaged small businesses. Pre-qualification data from 2004 and prime contract data drawn from the NJTA were evaluated. Using regression analysis, the levels of pre-qualification dollar amounts based on ethnicity, gender, tasks performed by the companies, and whether they are DBE or non-DBE firms were tested. Results revealed DBEs are more likely to be pre-qualified for contracts, but less likely to pre-qualify for high contract dollars.

Faculty Profile

Dr. Corliss W. Outley is currently an assistant professor in the Recreation and Sport Studies Division in the School of Kinesiology at the University of Minnesota. Dr. Outley attended Texas A&M University where she received her Ph. D. in Recreation and Resources Development in 2000. Her research specializes in the study of racial and ethnic patterns in recreation behavior, urban youth recreation and minority perspectives, and youth development among racial/ethnic groups. Dr. Outley is published in multiple research journals and has presented her work at conferences nationwide. Dr. Outley has been a McNair faculty mentor for the past five years.

Jessica A. Sawyer is a senior at the University of Minnesota, Twin Cities, double majoring in political science and sociology. Her research interests revolve around understanding access and retention issues impacting first-generation college students. Ms. Sawyer plans on getting her Ph.D. in Higher Education Policy and Administration.

Dream Statement

"My dream is to start my own non-profit organization and to receive a Ph.D. in the social sciences. I want to be involved in global conflict resolution efforts and I want to orate and write extensively."

_____ _____

Faculty Mentor Approval Date

Week 8

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Week 9

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Presenting your Research Poster Effectively

Posters are a great way to present information in a more informal setting than an oral presentation. Given that posters are one-on-one presentations, they allow visitors to chat with the presenter, to ask specific questions, make suggestions, and follow your results.

During the presentation, the author stands beside the poster and guides viewers through the presentation. People circulate from poster to poster asking questions and talking to the author. You will stand by your poster for up to two hours.

A visitor to a poster does not want to read it, but to inspect it. Thus, the key to creating an effective poster presentation is visual simplicity achieved without loss of information content. In terms of visual design of the poster, substance is more important than presentation

Poster sessions have become a significant communication style for presenting research results in professional association meetings in all disciplines of the natural and social sciences. Visitors to poster sessions see and hear more presentations than in the equivalent time dedicated to presented papers; moreover, the opportunity for close conversation between presenter and visitor is one of the most important ways that information can be shared among scholars.

Your job at the poster session is to let people read your poster and ask questions. Remember that people won't know what questions to ask until they have viewed your poster, so allow them some time before you engage them. You might engage people by explaining the basics of your research in a 10-20 second spiel. You might also ask if they have any questions, or explain information that other attendees have asked questions about.

Common questions asked by conference attendees:

-Tell me about your research
-Why are you interested in this topic?
-What were your methods in conducting this research?
-How long has this study been going on? (long or short term?)
-Is this a well-studied area?
-What are your recommendations for further research?

Adapted on 5/28/08 from
http://faculty.uccb.ns.ca/pmacintyre/apics/presentationtips.htm#POSTER and
http://www.aas.duke.edu/trinity/research/vt/postertips.html

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Week 10

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McNair Scholars 2012
Permanent Address Form

Name: _____

Permanent address:

_____

_____

_____

_____

Name and address of 2 relatives or friends who have resided at the same address for at least 5 years (for follow-up purposes):

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

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Self-Assessment Questionnaire (Posttest)
McNair Scholars Program

Please indicate your comfort level with aspects of the research process and applying to graduate school.

The Research Process	*Uncomfortable*	*Somewhat uncomfortable*	*Not sure*	*Somewhat comfortable*	*Very comfortable*
Writing skills (general)	1	2	3	4	5
Introduction/Hypothesis/ Statement of the problem/ Significance of study	1	2	3	4	5
Literature review/ Citation styles	1	2	3	4	5
Research Method	1	2	3	4	5
Results/Discussion/ Implications	1	2	3	4	5
Abstract	1	2	3	4	5
Talking to and collaborating with a professor	1	2	3	4	5
Oral presentation skills	1	2	3	4	5
Preparing a Poster on your research	1	2	3	4	5
Graduate School Preparation					
Knowing that I want to go to graduate school	1	2	3	4	5
Preparing a curriculum vitae	1	2	3	4	5
Writing a personal statement	1	2	3	4	5
Finding a suitable graduate school programs	1	2	3	4	5
Finding funding for graduate school	1	2	3	4	5
Taking standardized test (e.g. ACT, GRE, LSAT)	1	2	3	4	5

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McNair Scholars Summer Program Evaluation

Please complete the following program evaluation.
It will provide the staff with feedback that will help in planning future summer programs.

Expectations:

1. Did the program meet your needs and expectations?
(circle one) **YES** or **NO**

1a. What skills and/or knowledge did you hope to gain from participation in this program?

(please list)

_____

_____

_____

_____

_____

2. Please circle all seminars that you found helpful

Taking the Graduate Record Examination

Meeting with Graduate School Rep.

Effective Writing for Graduate School

Graduate School Research

Development of Personal Statement

PowerPoint Poster Training by UMN Librarian

2a. What type of seminars would be helpful that were not offered during the summer?

_____

_____

_____

_____

_____

Research Project:

3. Were you assigned a research project in your area of interest?
(circle one) **YES** or **NO**

4. Please indicate how many hours were spent on a weekly basis (on average) working on your research project? _____ hours

5. Please list any research skills that you learned this summer as a result of working on your project.

_____

_____

_____

_____

_____

Please circle the response that most applies:

6. I would rate my academic effort during the summer as (please circle):

harder than I've ever worked	**Equal to college classes**	**average**	**Not as hard as college**	**little effort**

7. I would rate the quality of my work as:

exceptional (my best ever)	**better than most academic work**	**average**	**below average**	**little effort**

8. Do you feel that your poster accurately reflected the amount of work that you spent on your research project?
(circle one) **Agree** or **Disagree**

Comments:

_____

_____

_____

_____

_____

Reflection:

9. Please describe how you anticipate your participation in McNair Scholars will affect your future academic performance. (selection of classes, selection of major, increased motivation to do well and study harder)

_____

_____

_____

_____

_____

10. How do you intend to utilize the services of this program after this summer?

_____

_____

_____

_____

_____

Faculty:

(please circle the response that most applies)

11. How many hours per week did you spend working with your faculty mentor?
_____ hours.
My mentor will: (circle one) **Agree** or **Disagree**

12. My faculty mentor was available and provided the necessary guidance for my research project.

(circle one) **Agree** or **Disagree**

13. I feel confident that the relationship with my mentor will continue after this summer.

Agree or **Disagree**

14. I feel confident that my mentor will write me a strong letter of recommendation.

Agree or **Disagree**

15. McNair staff provided guidance when problems surfaced with the faculty mentor or project.

Agree or **Disagree**

16. Checking in with McNair staff and McNair seminars provided academic and personal support.

Agree or **Disagree**

17. The best part of working with my faculty mentor was:

_____

_____

_____

_____

_____

18. I could have been more productive in my research if my faculty mentor had:

_____

_____

_____

_____

_____

19. I could have been more productive in my research if the McNair staff had:

_____

_____

_____

_____

_____

20. I could have been more productive in my research if I had:

_____

_____

_____

_____

Participation in the McNair Program has influenced my:
Please circle the response that most applies

21. Interest in attending graduate school

Increased **remained the same** **decreased**

22. Motivation to prepare academically for graduate school

Increased **remained the same** **decreased**

23. Confidence that I can be successful in graduate school

Increased **remained the same** **decreased**

24. Motivation to work harder during my undergraduate studies

Increased **remained the same** **decreased**

25. Ability to think critically (examine issues more thoroughly)

Increased **remained the same** **decreased**

Additional Comments:

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

Name (optional): _____

Date: _____

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University of Minnesota
McNair Scholars
Summer 2012

This is to certify that I have examined this copy
of the McNair research title and project
by

_____
(Student Name)

and have found that it is complete and satisfactory.

_____
Name of Faculty Mentor(s)

_____
Signature of Faculty Mentor(s)

_____/_____/_____
Today's Date

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